



POLITECNICO
MILANO 1863

FRECCIAROSSA 1000

**Technical and Economic Analysis of *Frecciarossa 1000*
on High-Speed Line Turin-Milan**

Electric Systems for Mobility

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Group 9

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Mechanical and Electrical report

We have chosen to analyse the path of *Frecciarossa 1000* between the railway station of Torino Porta Nuova and Milano Centrale.

Train description



Figure1.

Frecciarossa 1000.

The commercial name of the electric train is *Frecciarossa 1000* (also known as ETR 1000) and it is an EMU (electric multiple unit) manufactured by Hitachi Rail Italy (Ansaldo Breda) and Bombardier, while the design part was developed by Bertone.

The ETR 1000 is a bidirectional train with locked composition and symmetric architecture: there are 2 independent semi-train for a total of 8 coaches. Each coach has 2

bogies and each bogey has 2 wheelsets: the total number of wheelsets is 32 but the traction is distributed and the motorized wheelsets are 16. For each motorized wheelset there is a three – phase asynchronous motor, the total number of engines is 16. The wheel arrangement is the following: Bo'Bo'+2'2'+Bo'Bo'+2'2' +2'2'+Bo'Bo'+2'2'+Bo'Bo'. In normal condition the total length of the train is 202 meters with 455 seats + 2 tip-up seats; it is even possible to arrange two train in series reaching a total length of 400 meters with more than 900 seats.

This train is approved and takes service in Italy but it is prepared for operations in other 7 European countries; indeed it is equipped to be supplied by different lines: AC 25 kV, 50 Hz & DC 3 kV (base); DC 1,5 kV & AC 15 kV, 16.7 Hz (optional).

Some other technical information: the top reachable speed is 400 km/h, but the top commercial speed is 360 km/h, the acceleration on departure is $0,7 \text{ m/s}^2$. Other technical aspects are reported in the following table:

Gauge	1435 mm
Width of coaches	2924 mm
Height of coaches	4080 mm
Wheelbase of the bogey	2850 mm
Mass during service	500 t
Rotating mass	20,6 t
β	0,04
Maximum power to the wheels (DC 3 kV)	6,9 MW
Maximum power to the wheels (AC 25 kV)	9,8 MW
Maximum braking power (DC 3 kV)	5,65 MW
Maximum braking power (AC 25 kV)	10,5 MW
Power absorbed by auxiliary services	220 kW
Maximum braking deceleration (DC 3 kV)	$-0,48 \text{ m/s}^2$
Maximum braking deceleration (AC 25 kV)	$-0,58 \text{ m/s}^2$
Tractive efficiency	0.9
Braking efficiency	0.9

Table 1.

Braking phase

Frecciarossa 1000 has a particular braking system: the braking phase is performed using both pneumatic brake (dissipating mechanical power) and electric brake (converting mechanical power in electric power). The converted electric power can be dissipated by rheostats or transferred to be stored or used. The special behaviour is really about the electric brake: only in few cases you can recover the electric power converted from mechanical power during the braking phase, most of the times the converted power is dissipated. As we can see from the following graphs, in order to recover high amount of electric power we have to brake requesting a little percentage of the total braking power and that means we have to start braking very far away from our target position. This is a big limitation if we want to recover energy in real life operations where we want the travel time to be as short as possible.

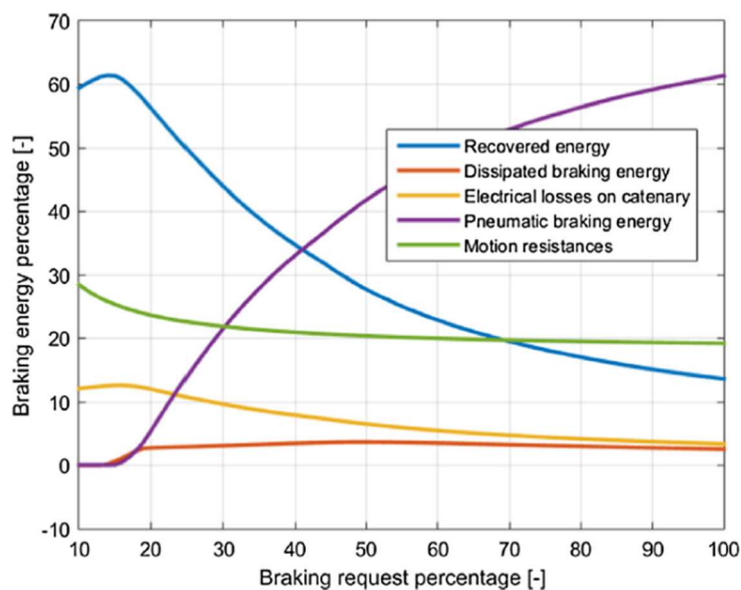


Figure 2.

Energy percentages involved in the train braking phase (measured at the pantograph) as a function of the braking effort percentage, voltage peak limited to 3900 V.

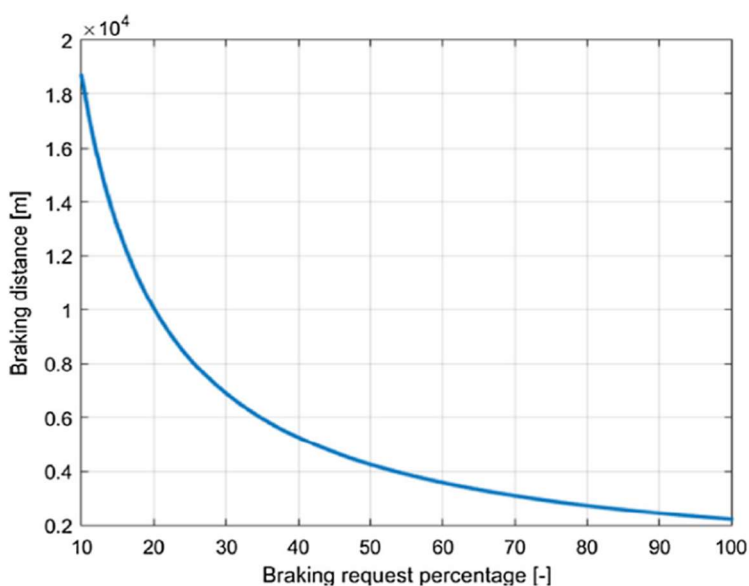


Figure 3.

Braking distance as a function of the braking effort percentage, without voltage limitations.

Geographical features of the path

The geographic coordinates have been obtained from a research setting the starting and ending station and selecting High-Speed routes on the web site “Railway Routing” (<https://signal.eu.org/osm>). Then we imported the results on Google Earth Pro to visualize and extract more information about length, slope and curves. The total length of the path is 148 km, with two intermediate stops: one after 5 km from the beginning corresponding with the Torino Porta Susa station and the other after 135 km corresponding with the Rho Fiera station.

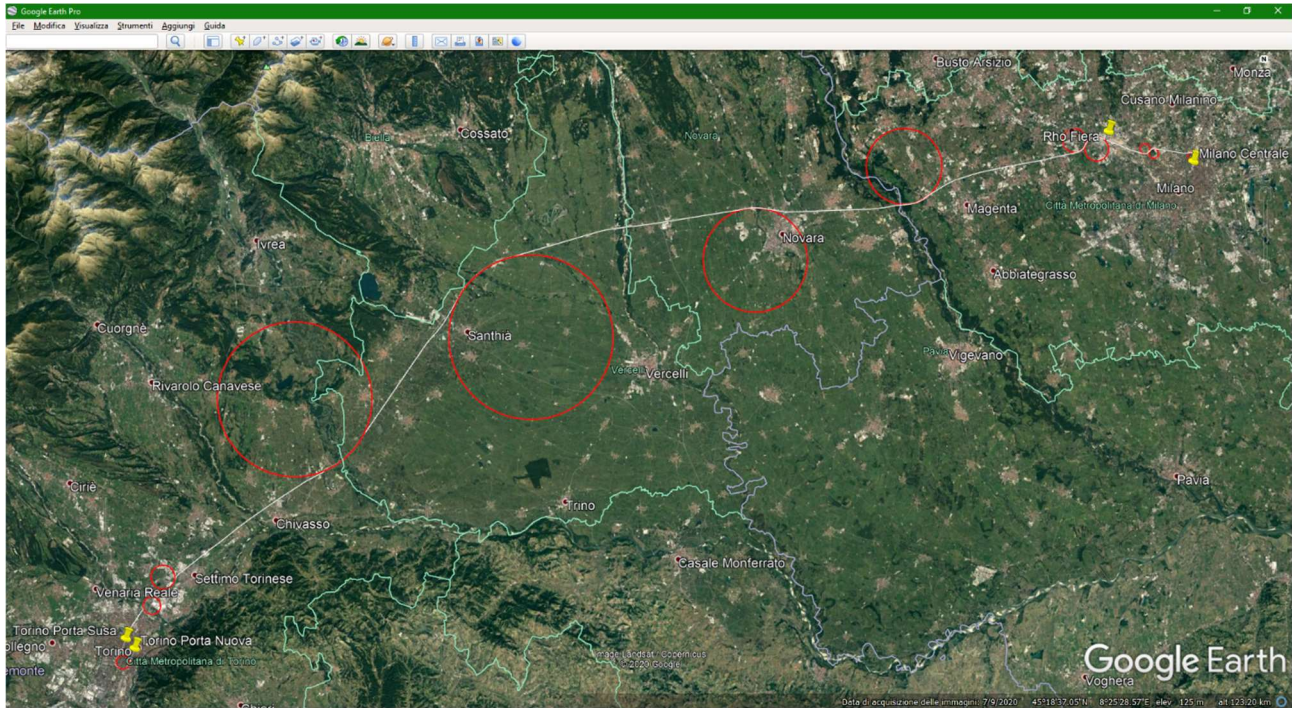


Figure 4.

The railway path with stops and curves.

We created a simplified model of the path in *MatLab* starting by the data from Google Earth, indeed we built a model assuming only straight lines and twelve curves, identified among the most significant ones. The parameter that we considered for the definition of the curves were radius, starting point of the curve and ending point. In this way we were able to compute the resistance due to the curvature, identify where to consider this resistance in the calculations and for how many meters it was present.

	Radius [m]	Starting Point [m]	Ending Point [m]
curve 1	770	1070	2860
curve 2	830	9930	10800
curve 3	1330	13300	14600
curve 4	8530	35000	41000
curve 5	9040	59400	67200
curve 6	5750	93200	95700
curve 7	4240	111000	114000
curve 8	1260	131000	133000
curve 9	1340	133000	135000

curve 10	540	140000	141000
curve 11	540	141000	142000
curve 12	510	145000	147000

These curves have a wide range of the curvature radius because they are located in two different parts of the railway path with different speed limits: the path supplied by 3 kV DC and the path supplied by 25 kV AC.

The mean value of the slope for the entire path is - 0.4%. For the calculations we considered pieces of 10 km with a constant value of slope in ‰.

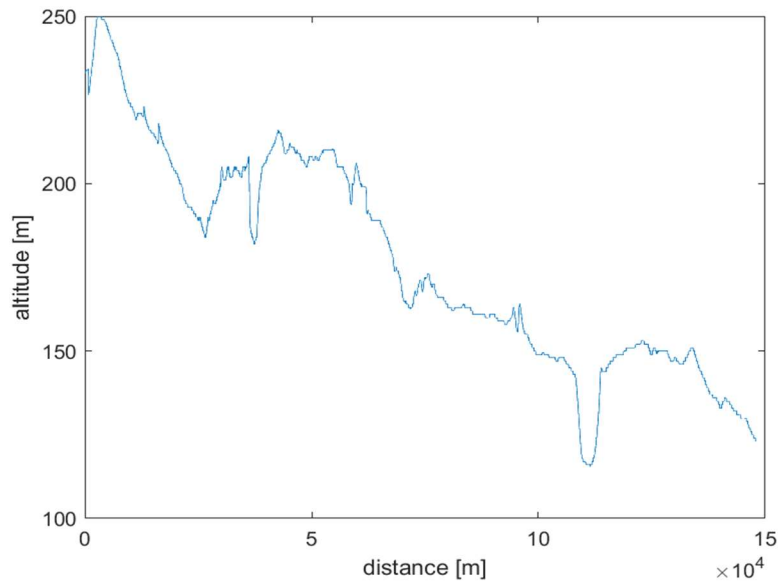


Figure 5.

Elevation profile of the path.

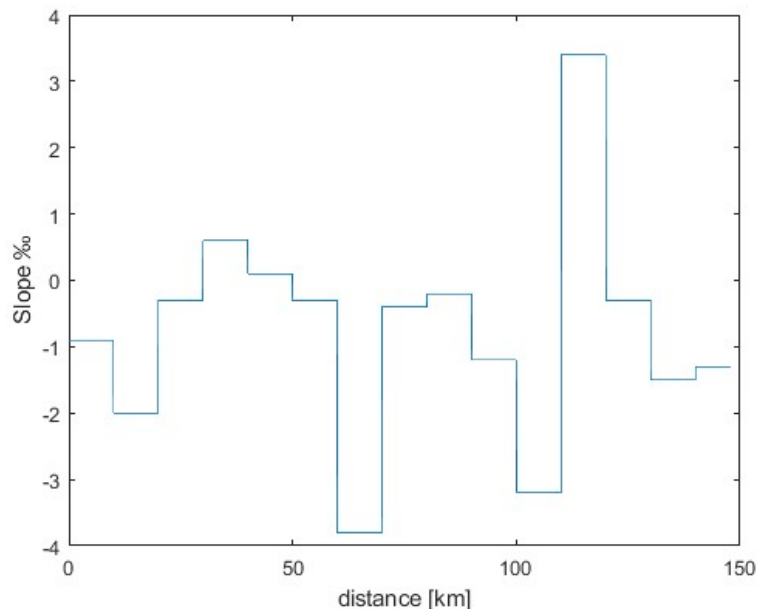


Figure 6.

Approximated slope of the path.

For the definition of the speed limits on the various sections of the path and to identify in which points the 3 kV DC supply ends and the AC supplied section starts we used another web site called “OpenRailwayMap” (<https://www.openrailwaymap.org/>).

Matlab Simulation

Characteristics of the line

Figure 7 shows the track considered for the simulation; the coloured dots indicate the positions of the train stops. Figure 8 reports the main characteristics of the line in terms of speed and stops.

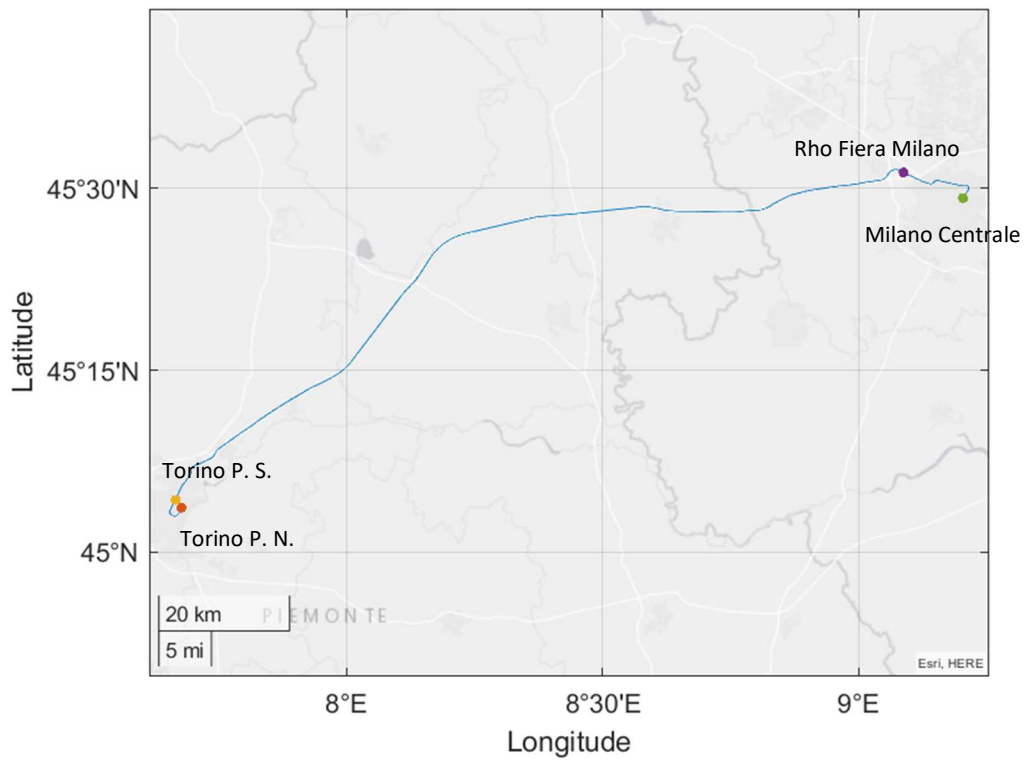


Figure 7.

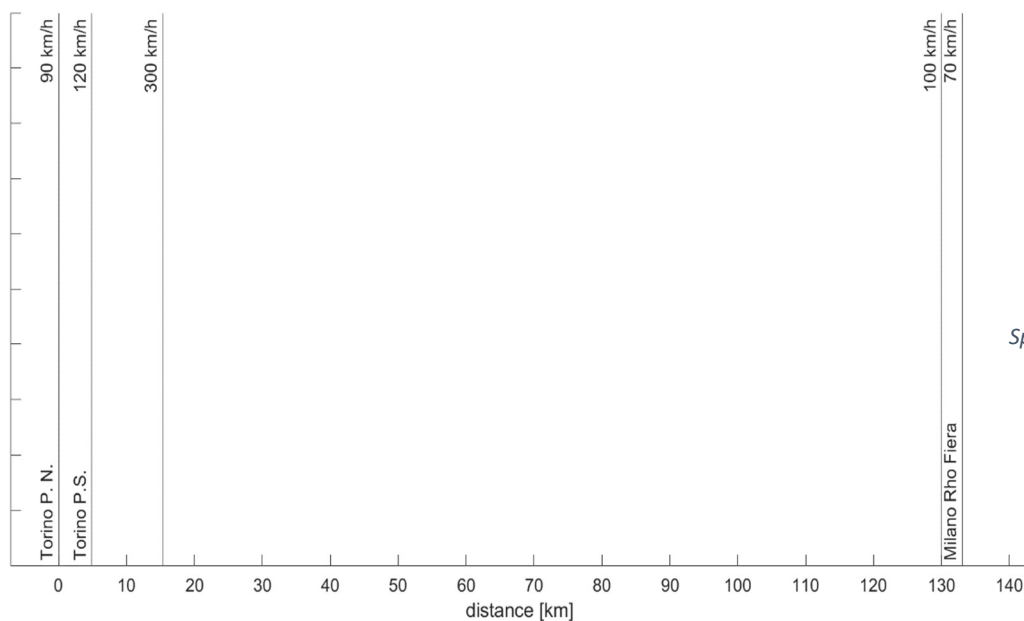


Figure 8.

Speed limits starts and train stops considered in the simulation.

The borders between AC and DC lines (EBP, Electric Border Point) are assumed 1.6 km long and located as indicated in the following

$$EBPs: \begin{cases} 11.640 \text{ km} \leq EBP_1 \leq 13.240 \text{ km} \\ 131.000 \text{ km} \leq EBP_2 \leq 132.600 \text{ km} \end{cases}$$

EBP_1 is just before 300 km/h speed limit start point; EBP_2 is considered to start at 100 km/h limit.

The EBPs must be traveled to low pantographs so the current absorbed by the train and the electric power supply are zero in these sections.

Results

Table 3 reports on the left the time schedule of a journey took from Trenitalia website. We also reported our results from our simulation. When the train reaches the station a stop of 2 minutes is considered as for the real case. We suppose the train leaves at time equal to 0. The right part of the table shows the difference in terms of distance between the points where the train was supposed to stop and those obtained with the simulation.

Station	Arrive Time [minutes]		Stop point [m]		Difference [m]
	Real journey	Results	Supposed	Results	
Torino Porta Susa	8	7	4870	4886	16
Rho Fiera Milano	44	42	133035	133084	50
Milano Centrale	60	58	147878	147909	32

Table 2.

Dynamic quantities

In this section we report the results for acceleration, speed and space. In the graphs braking phases and the EBPs are highlighted with red and yellow dots.

To calculate such quantities the following formulas were used:

$$a = \frac{F - R}{m_e}$$

$$v = a\Delta t + v_0$$

$$s = \frac{1}{2}a\Delta t^2 + v_0\Delta t + s_0$$

where F is the force applied by the train, R is the overall resistance and m_e is the equivalent mass calculated as $m_e = m(1 + \beta)$.

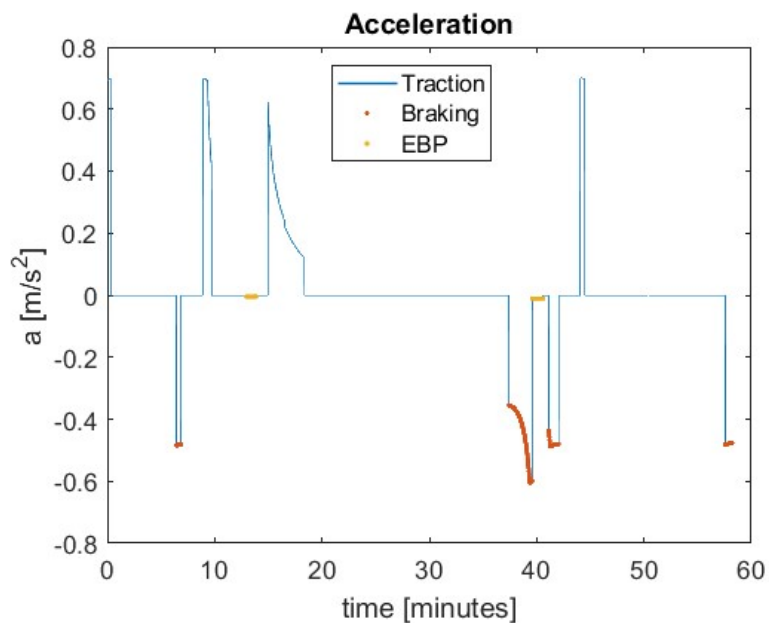


Figure 9.

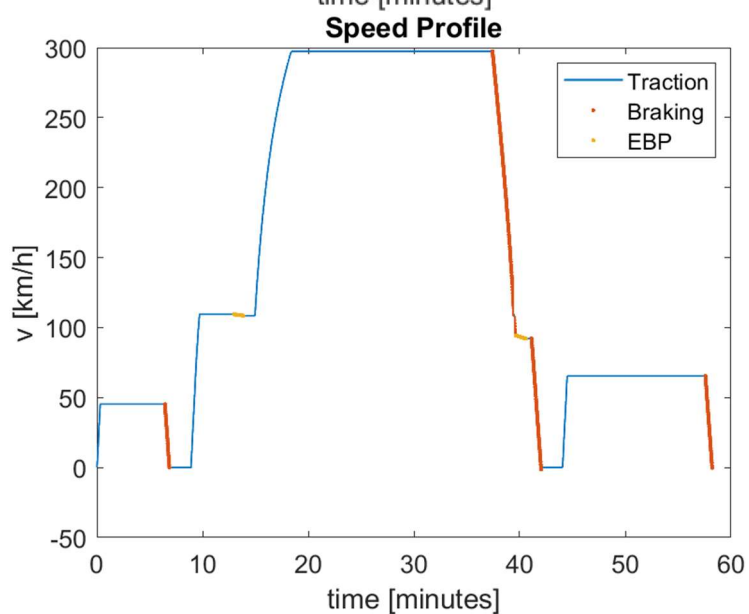


Figure 10.

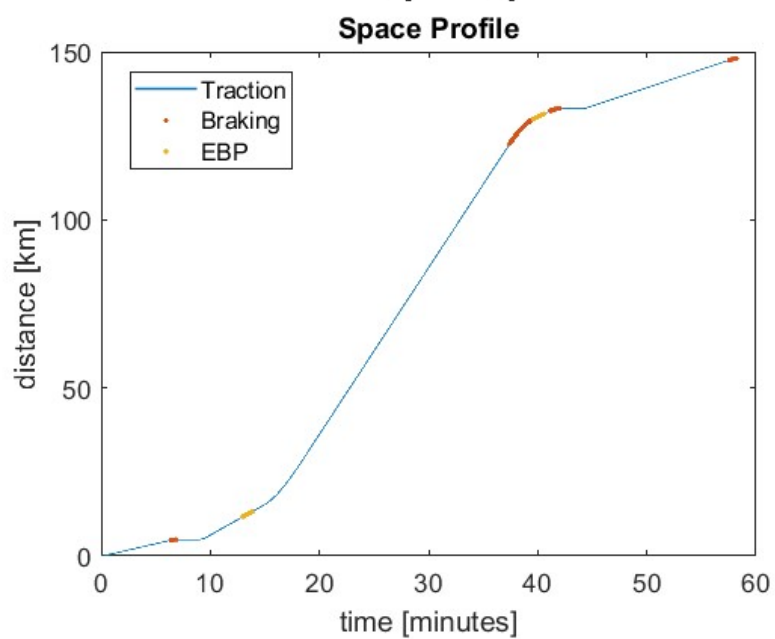


Figure 11.

Traction, braking and resistances

As can be noticed from the pictures in the free running condition the tractive force is equal to the overall resistance. At a constant speed of 300 km/h the air resistance is constant since it depends just on the speed. As a result of the constant speed the tractive force follows just the trends of slope and curvature resistance as can be seen in Figure 13.

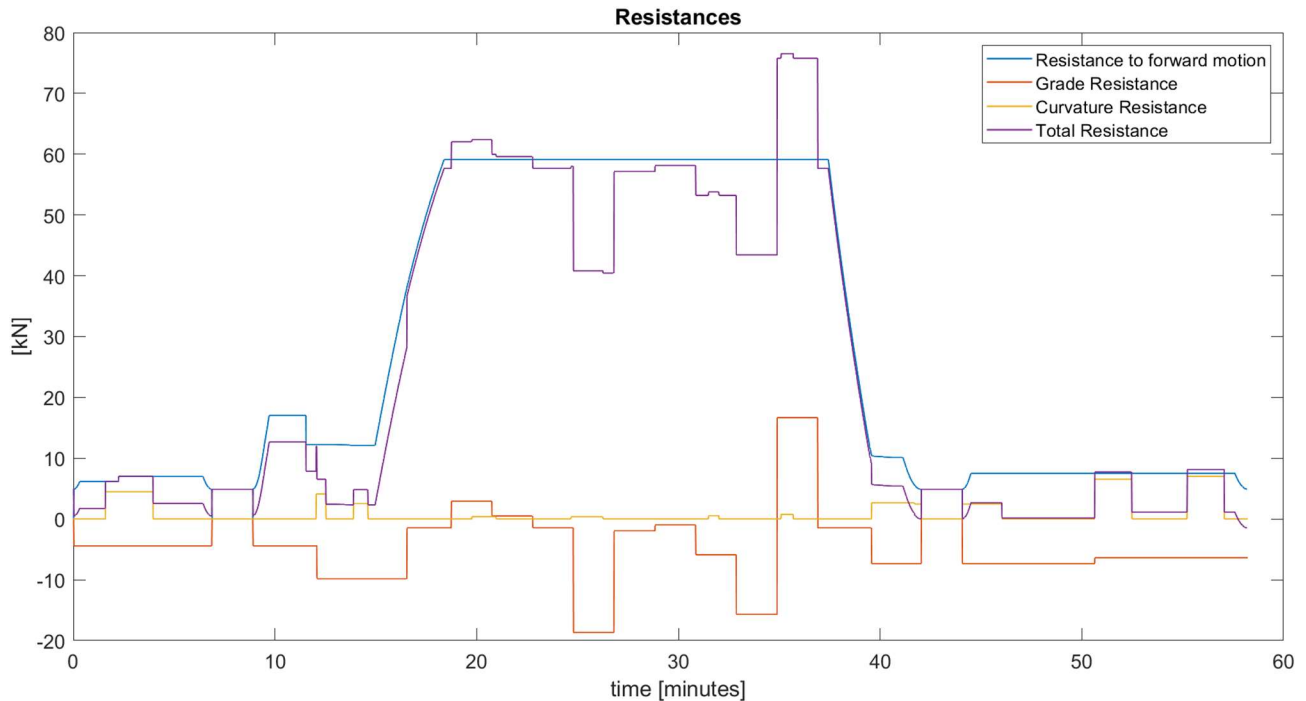


Figure 12.

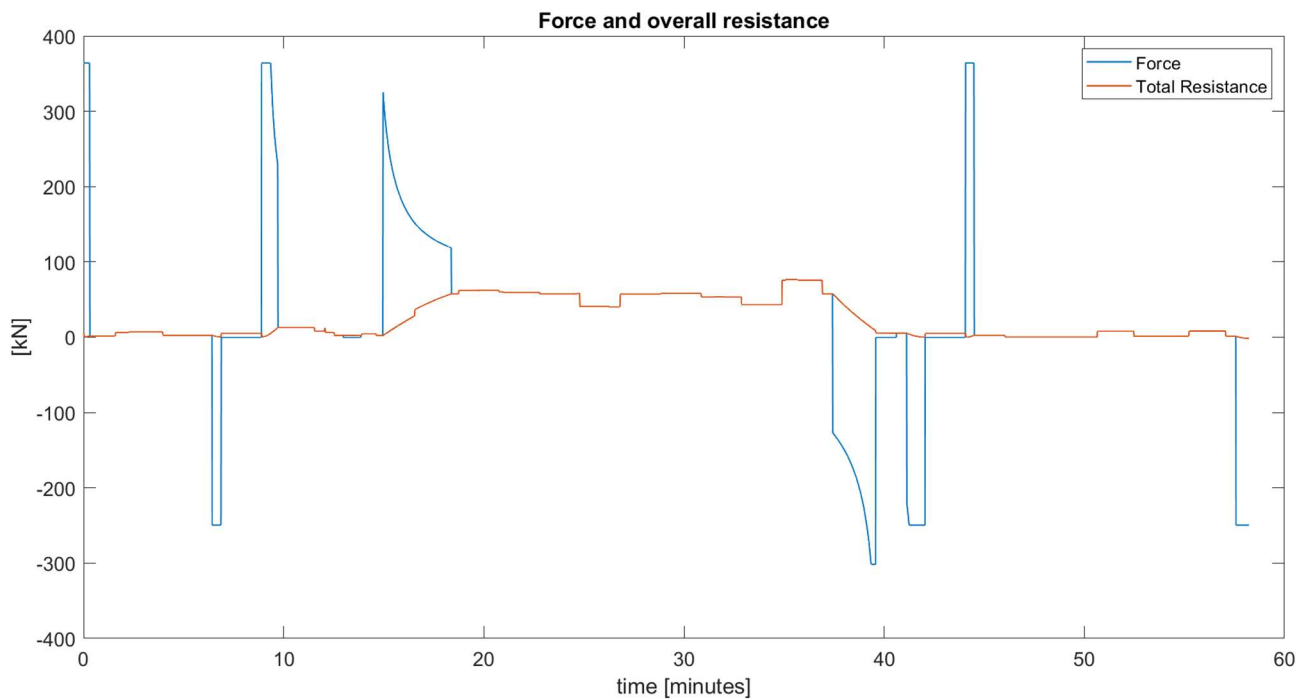


Figure 13.

Resistances are calculated as follow:

Resistance to forward motion

$$R_o[N] = \begin{cases} \left(1 + 0.0125 \left(\frac{v}{10}\right)^2\right) mg, & \text{in open air} \\ \left(1 + 0.0207 \left(\frac{v}{10}\right)^2\right) mg, & \text{in tunnel} \end{cases}$$

Where v is the speed expressed in km/h, m is the mass of the train in tons.

We considered the presence of a tunnel section in the urban area of Turin between the track's points 1570 m and 9030 m.

Curvature resistance

$$R_c [N] = r_c mg$$

where $r_c = \frac{a}{\rho - b}$ is the curvature resistance coefficient (ρ = radius of curve), m is the mass of the train in kg.

Grade Resistance

$$R_i [N] = r_i mg$$

where r_i is the curvature resistance ad is assumed $r_i = i_{\text{‰}}$, m is the mass of the train in tons.

The overall resistance is computed as the sum of the above resistances and includes also the contribute of the rolling resistances calculated as $0.001 \cdot mg$.

$$R = R_o + R_c + R_i + 0.001 \cdot mg$$

Tractive and braking effort are computed based on the following trends for the ETR 1000.

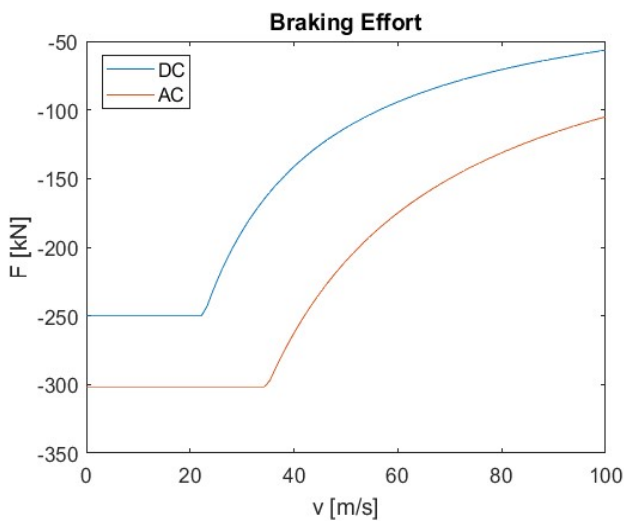


Figure 14.

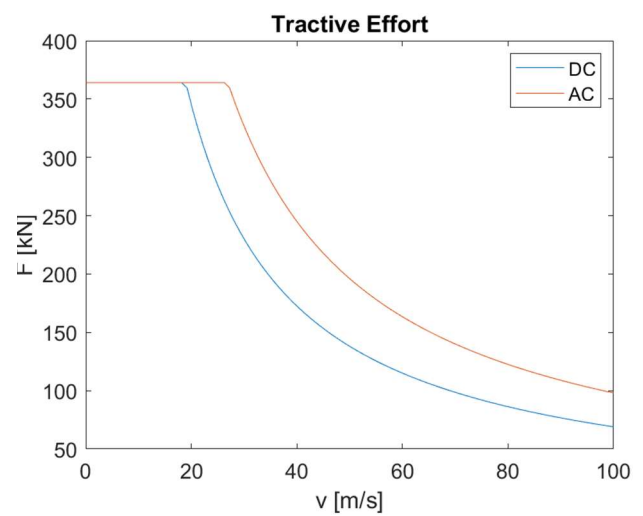


Figure 15.

Electrical quantities

Electrical quantities were calculated with the following formulas:

Electric power

$$P_{el} [kW] = \begin{cases} \frac{F \cdot v}{n_t} + P_{aux}, & \text{for traction} \\ F \cdot v \cdot n_b \cdot B_{el} + P_{aux}, & \text{for breaking} \end{cases}$$

F is the force applied by the train expressed in kN ;

v is the speed expressed in m/s ;

n_t is the tractive efficiency;

n_b is the braking efficiency;

B_{el} is the percentage of regenerative braking (between 0 and 1);

P_{aux} is the power of auxiliaries expressed in kW .

Current

$$I [A] = \frac{P_{el}}{V}$$

where P_{el} is expressed in kW and V (voltage line) is expressed in Volt.

Energy

$$E_{el} [kWh] = P_{el} \Delta t + E_{el0}$$

Figures 16, 17 and 18 show their trend during the entire journey. Braking and EBPs are highlighted in red and yellow dots.

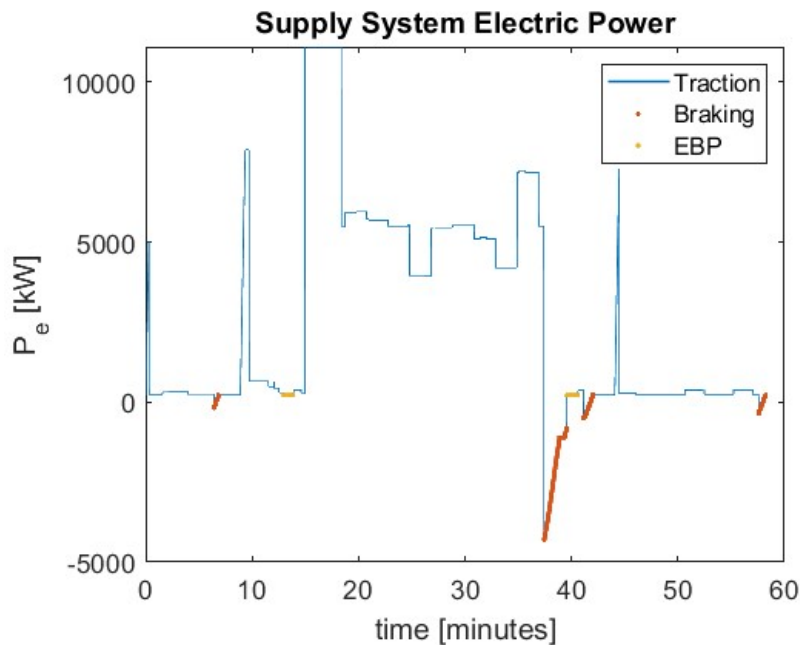


Figure 16.

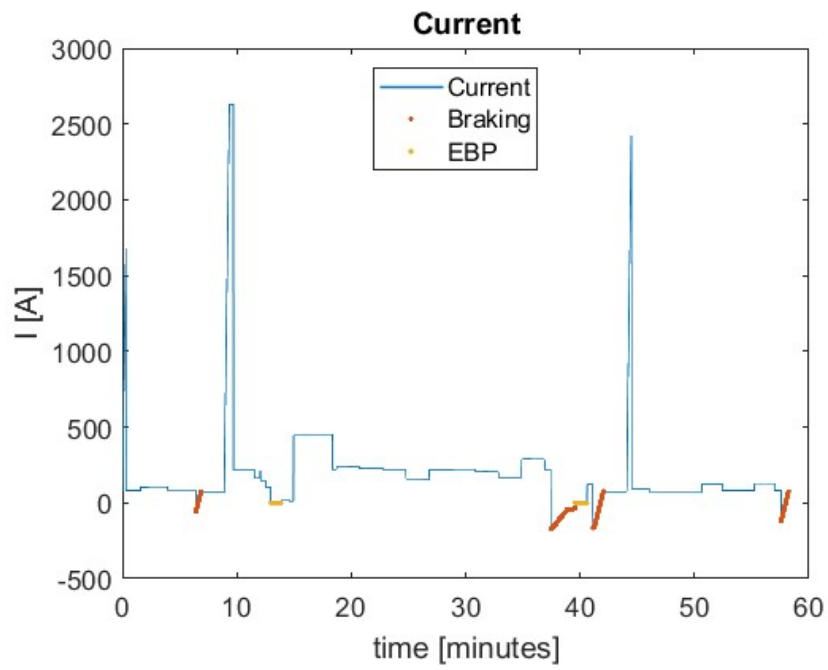


Figure 17.

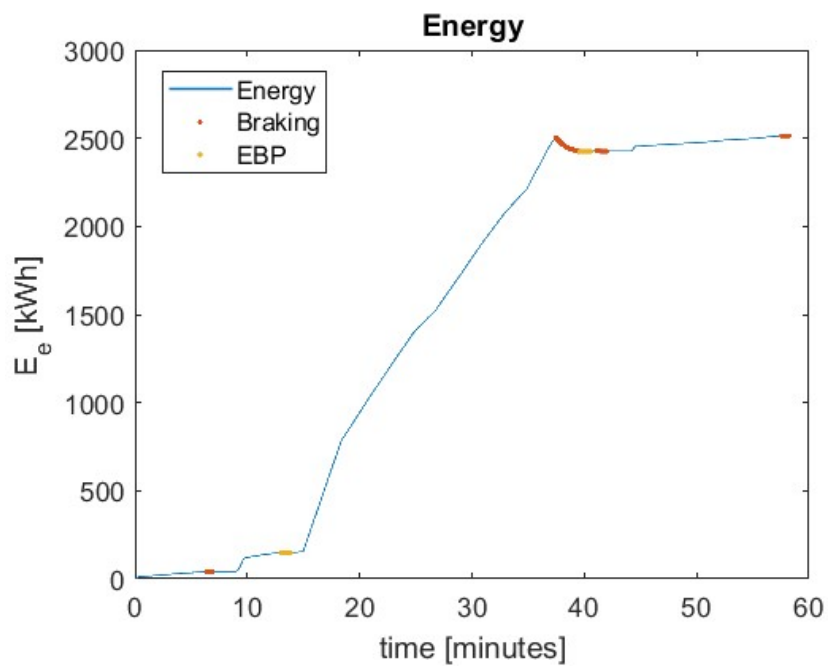


Figure 18.

To check the non-slip condition, we analyzed the tractive and braking effort as a function of the train speed. In red all the efforts (tractive Figure 19, braking Figure 20) registered during the entire journey. In blue the corresponding adhesion force for the same speed. As can be noticed from the graphs tractive or braking forces are never over the adhesion limit.

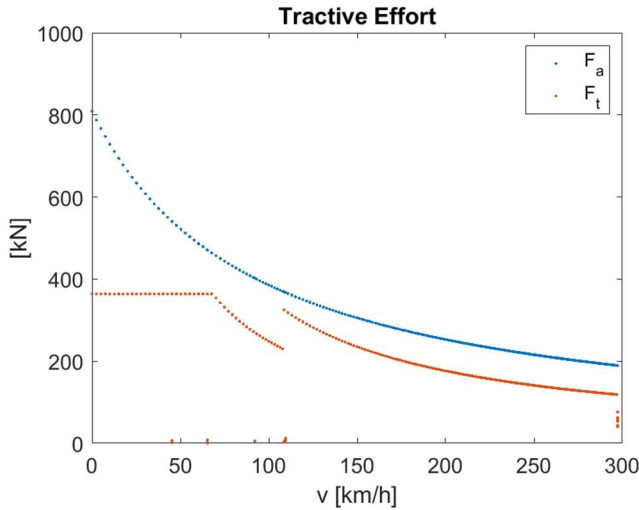


Figure 19.

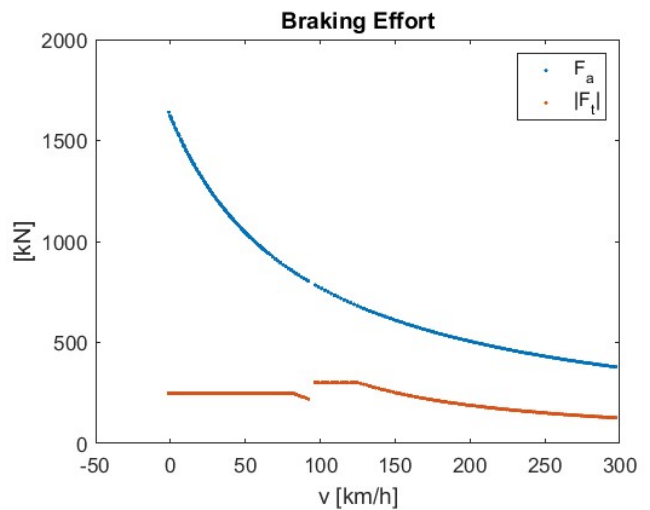


Figure 20.

With regards to the braking, we took into account Figures 2 and 3 for the particular behavior of ETR 1000. For each braking distance calculated we estimated the braking request percentage from Figure 21 and then according to Figure 22 we calculated the Braking Energy Percentage in order to compute the negative electric power. Such electric power converted during braking phase is generally not recovered but dissipated by rheostats.

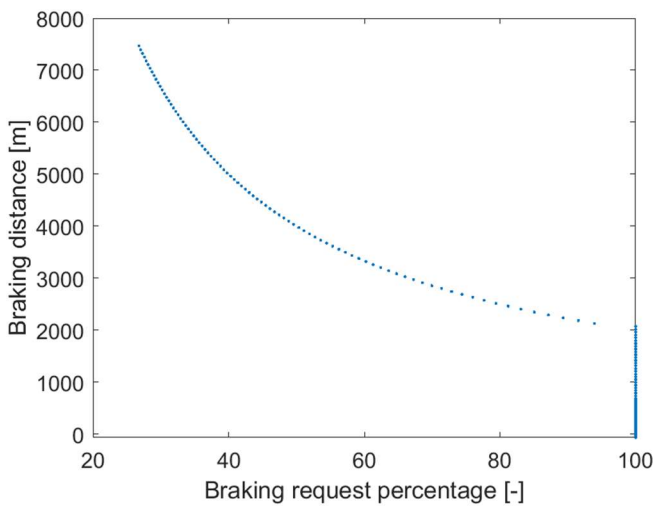


Figure 21.

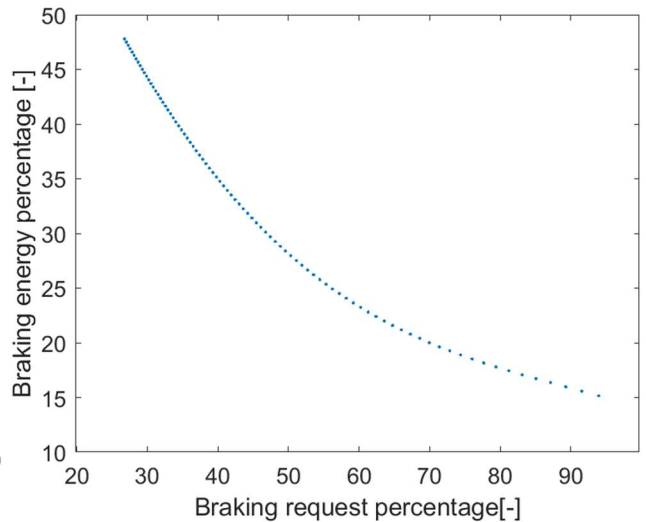


Figure 22.

Financial Analysis

The second part of our report deals with an economical and financial analysis on High-Speed trains fleet employment as a consequence of the Covid-19 spread all over the World.

Financial Data and hypothesis

The pandemic affected the whole global economy and transportation and mobility systems have registered dramatic losses. In particular, Railway Transport productivity relies on the huge crowd demand and shows a significant damage. With regards to Italy situation, FSI (Ferrovie dello Stato Italiane) Balance Sheet for the first semester of 2020 presents an important downfall of all the income indexes: -1122 M€ of gross operating income, -600 M€ of operating income for Long Haul, EBITDA = 518 M€ in opposition to 1297 M€ of first half of 2019 and 1265 M€ estimated without Covid-19.

After the study of the FSI's Balance Sheet and Income Statement for the years 2018-2019-2020, we tried to estimate the average exercise cost for High-Speed trains because the detailed cost allocation is not available. We started from the total kilometers travelled every year ($L = 67224000 \text{ km} * \text{train}$) and the mean number of daily *Frecciarossa* runs (between 275 and 290). The Long Haul (LH) operating revenues for the first semester of 2019 and 2020 amount to 1273 M€ and 573 M€, with a decrease of 55% due to -620 M€ from *Frecciarossa* and -80 M€ from *Intercity* ticket sales. The operating revenues and costs from transport activities for 2019 were 8160 M€ and 6278 M€.

These revenues rose of 230 M€ from 2018 to 2019, 85 M€ of which from LH transport. An approximate estimation of *Frecciarossa* costs and revenues gave us:

$$TR_{2019} = \frac{85}{230} * \frac{620}{700} * 8160 = 2671 \text{ M€}$$

$$TC_{2019} = \frac{85}{230} * \frac{620}{700} * 6278 = 2055 \text{ M€}$$

$$m = \frac{2055 \text{ M€}}{62,774 \text{ Mkm} * \text{train}} = 30,6 \frac{\text{€}}{\text{km} * \text{train}}$$

In our opinion this value was unrealistic because of the excessive approximation, but it gave us the correct magnitude of unitary cost. We will focus on the economical sustainability of HS after explaining the model we developed.

The poll

We decided to make a poll in order to estimate how the Covid-19 pandemic has influenced the habits of the people. The poll's final aim was to find out the general lack of trust in public transport safety. In order to get a more balanced feedback, we divided the questions depending on their aim: profiling the respondents for the first seven and trying to predict their behavior with the last three of ten.

The profiling part deals with the main features of the respondents: the frequency of railway transport, the average price, length and duration for their trips, the crowding of the train. We cannot report Question #6, “Qual è la tratta che percorrevi più spesso? E il prezzo medio del biglietto?”, because it is open and we used its results to evaluate how well we spread our sample.

We used this information in order to weight their contribution differently, because we were more interested in who is more likely to take a train, in particular HS. We set these coefficients to lower the importance of the less frequent travelers (Tables 4 and 5).

Quale mezzo di trasporto utilizzavi più di frequente sulle lunghe distanze prima del COVID?
209 risposte

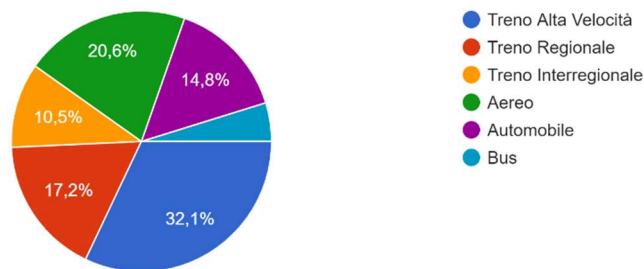


Figure 23.

Quanto spesso utilizzavi il trasporto ferroviario prima del COVID? (in ambito extraurbano)
209 risposte

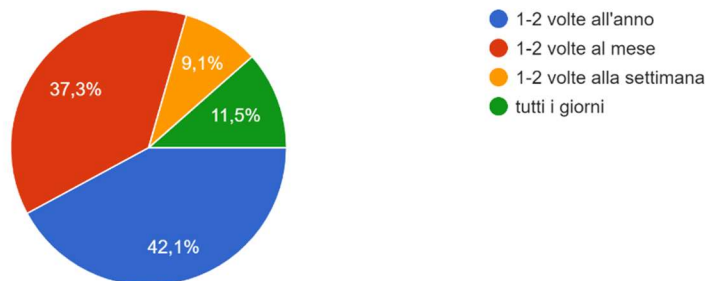


Figure 24.

Quale tipo di trasporto ferroviario utilizzi più spesso? (in ambito extraurbano)
209 risposte

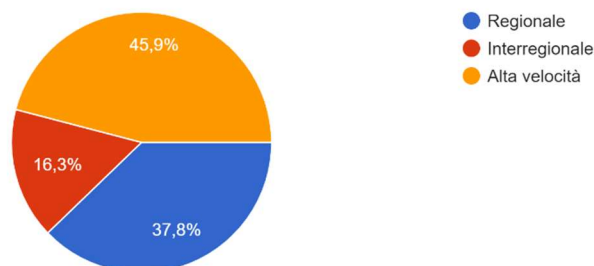


Figure 25.

Quanti chilometri percorrevi in media prima del COVID con il trasporto ferroviario? (in ambito extraurbano; media per viaggio)

209 risposte

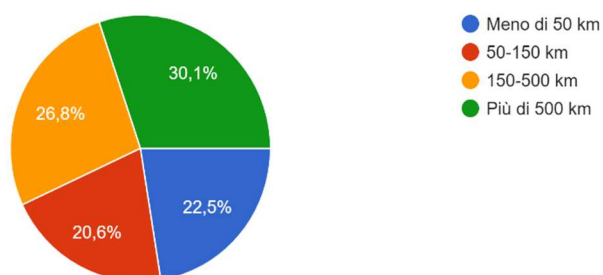


Figure 26.

Quanto tempo impiegavi mediamente per i tuoi spostamenti? (Media per viaggio)

209 risposte

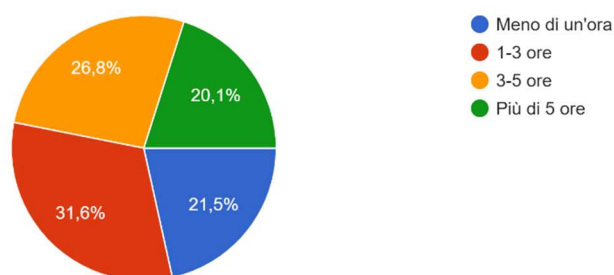


Figure 27.

Indica l'affollamento medio del tuo vagone su questa tratta. (1: la maggior parte dei posti vuoti; 5: tutti i posti occupati ed eventualmente alcuni passeggeri in piedi)

209 risposte

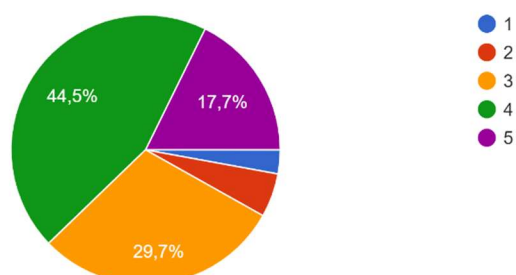


Figure 28.

Ferroviario (Regionale, Interregionale, AV)	Non Ferroviario (Bus, Aereo, Automobile)
1,00	0,50

Table 4. Weights for question #1.

Regionale	Interregionale	Alta velocità
0,25	0,50	1,00

Table 5. Weights for question #3.

The prediction part tries to understand the possible demand for 2021, both in case the virus maintains its danger and in case governments manage to limit it. The last question helped us to estimate the impact of capacity on HS demand; we developed our Reluctance model based on these answers.

Nel 2021 quale mezzo di trasporto privilegerai per le lunghe distanze?

209 risposte

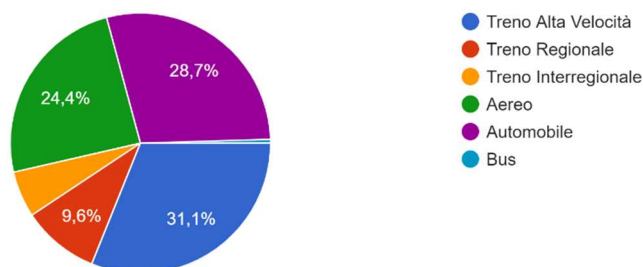


Figure 29.

E nel caso di contenimento efficace del virus? (a.e. crollo della curva dei contagi, disponibilità ed efficacia del vaccino, ecc.)

209 risposte

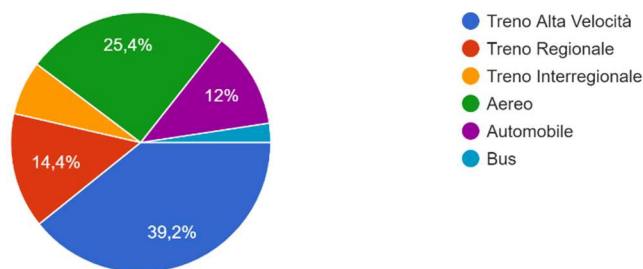


Figure 30.

Nel caso di treni ad Alta Velocità, con quale percentuale di capienza massima acquisteresti il biglietto?

200 risposte

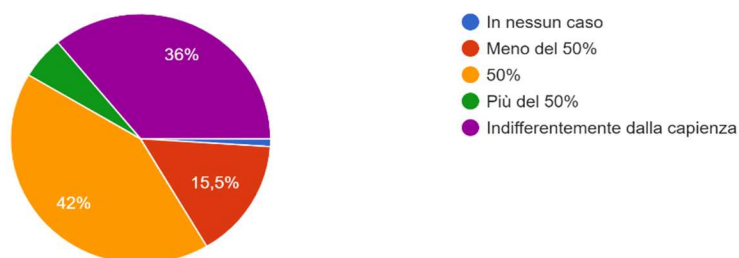


Figure 21.

There are several interesting consequences. First of all, the increased fear and mistrust could lead to an important rise of cars employment and therefore of related pollution. In case of significant protection and prevention measurements against Covid-19, the percentage of cars drops from 28,7% to 12%.

We used Python for a quick data analysis to underline relationship between customers trends. For example, Figure 32 shows the weighted number of Equivalent Passengers of HS follows a distribution resembling the last poll pie chart. This justifies our assumption of directly considering the poll data for our model.

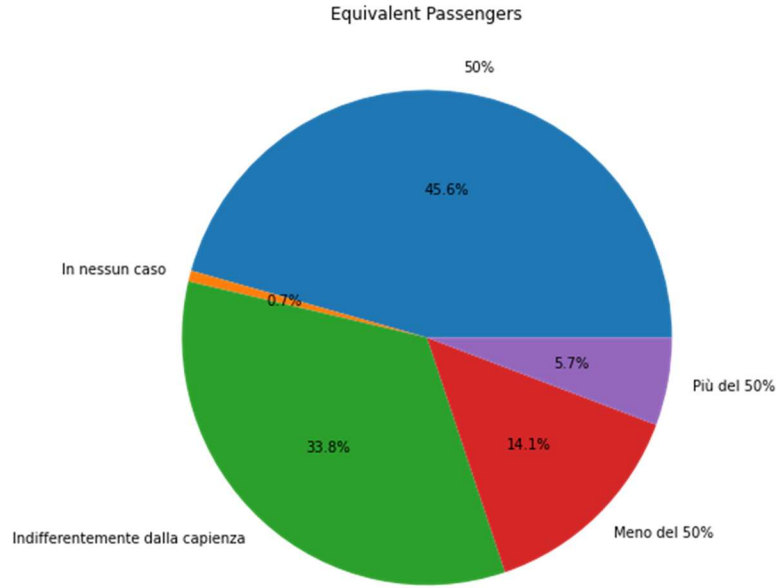


Figure 32.

The model

In order to face the pandemic and its consequences on economy, railway transportation companies had to increment travel tariffs to compensate the smaller capacity. This had an indirect fallout on the customers' willingness, who had to decide if that run was worth of a higher price. With the aid of poll data, we tried to develop a model for the effectiveness of ticket value rises.

The equation we started from is:

$$C_t = R_{TV} / N_{PT},$$

where R_{TV} is the total revenues for the run, N_{PT} is the number of passengers, $C_t \geq 0$ is the average cost of ticket.

Our first assumptions are $N_{PT} = N_{PT}(\mathcal{R}, \mathcal{C}, NS)$, $\mathcal{R} = \mathcal{R}(\mathcal{C}, C_t, L, T)$, where:

- $NS \geq 0$ is the number of seats in standard conditions for a HS train (for ETR 1000, 450 seats);

- $\mathcal{C}, 0 \leq \mathcal{C} \leq 1$, is the maximum allowed capacity, a percentage decided by government and authorities (1 before pandemic, 0.8 for public transport and 0.5 for LH after Covid-19 spread in Italy);
- $\mathcal{R}, 0 \leq \mathcal{R} \leq 1$, is the *reluctance*, a variable that tries to quantify the lack of attractiveness of a train run depending on the crowding, the cost C_t , the length L and duration T . $\mathcal{R} = 0$: the customer will always buy that ticket regardless of the cost and number of other passengers; $\mathcal{R} = 1$: the customer will never buy the ticket.

Therefore, we formulated some hypothesis on these variables, in order to simplify the model and manage to compute our financial results.

- $\frac{\partial \mathcal{R}}{\partial C_t}, \frac{\partial \mathcal{R}}{\partial T}, \frac{\partial \mathcal{R}}{\partial L} \geq 0$. The reluctance grows with cost even in normal conditions ($\mathcal{C} = 1$, no pandemic), because a higher cost is less attractive; at the same time, $\mathcal{R}$ is higher on longer trips, as underlined in the poll.

Lunghezza	Capienza Desiderata	N. risposte
150-500 km	50%	19
	Indifferentemente dalla capienza	16
	Più del 50%	4
	Meno del 50%	2
50-150 km	50%	2
Meno di 50 km	50%	2
	Indifferentemente dalla capienza	1
Più di 500 km	50%	26
	Indifferentemente dalla capienza	13
	Meno del 50%	9
	Più del 50%	2

Table 3. Length and desired capacity for HS.

Durata	Capienza Desiderata	N. risposte
1-3 ore	50%	9
	Indifferentemente dalla capienza	6
	Più del 50%	3
	Meno del 50%	1
3-5 ore	50%	22
	Indifferentemente dalla capienza	16
	Meno del 50%	6
	Più del 50%	2
Meno di un'ora	50%	2
	Indifferentemente dalla capienza	1
Più di 5 ore	50%	16
	Indifferentemente dalla capienza	7
	Meno del 50%	4
	Più del 50%	1

Table 4. Duration and desired capacity for HS.

Capienza Desiderata	Affollamento	N. Risposte
50%	4	27
	3	14
	5	8
Indifferentemente dalla capienza	4	17
	3	9
	5	2
	1	1
	2	1
Meno del 50%	3	6
	4	4
	5	1
Più del 50%	3	3
	4	3

Table 5. Desired Capacity and crowding for HS.

It is consistent with real prices and train performances that C_t, L and T are related in linear relationships with each other: $C_t \simeq C_0 + C_1 * L, T \simeq T_0 + T_1 * L$. Then, from now on we will consider only the ticket cost for the reluctance.

- $\frac{\partial \mathcal{R}}{\partial \mathcal{C}} \geq 0$. The reluctance grows with the capacity because people's mistrust increases with the number of available seats.
- $\frac{\partial C_t}{\partial \mathcal{C}} \leq 0$. Of course, a smaller capacity means that prices rise.
- In conclusion, $\mathcal{R} = \mathcal{R}(\mathcal{C}, C_t)$, $0 \leq \mathcal{R} \leq 1$.

We considered other reasonable properties of this 2-variable function.

- $\mathcal{R}(\mathcal{C}, 0) = 0$: an interested customer will always buy a free ticket!
- $\lim_{C_t \rightarrow \infty} \mathcal{R}(\mathcal{C}, C_t) = 1$: this ideal result is the natural consequence of $\mathcal{R} \leq 1$. We reach the maximum reluctance when nobody can practically buy the ticket.
- C_t is a mean price despite of the travel class or the time we buy the ticket. We therefore have to consider a threshold for a reasonable ticket, C_{MAX} (e.g. €160 for Rome-Milan, €75 for Turin-Milan). These are the only factors that are influenced by the start and end station: indirectly, C_{MAX} and mean values of C_t depend on the trip.

The result we proposed is a simple equation taking into account all of these properties.

$$\mathcal{R} = \mathcal{C} * \left(1 - e^{-\frac{C_t}{C_{MAX}}}\right)$$

An immediate outcome of this formula is that $\lim_{C_t \rightarrow \infty} \mathcal{R}(\mathcal{C}, C_t) = \mathcal{C}$ and $\mathcal{R} \leq \mathcal{C}$.

We can now give a practical meaning for the reluctance: $\mathcal{R}/\mathcal{C}$ is the most probable number of unsold seats on those available. It is possible to refine the model including a discontinuity corresponding to the threshold:

$$\mathcal{R}(\mathcal{C}, C_t) = \begin{cases} \mathcal{C} * \left(1 - e^{-\frac{C_t}{C_{MAX}}}\right), & C_t \leq C_{MAX} \\ \mathcal{C}, & C_t > C_{MAX} \end{cases}$$

Under the cost threshold, $\mathbb{E}[N_{PT}] = NS(\mathcal{C} - \mathcal{R}) = NS * \mathcal{C} * e^{-\frac{C_t}{C_{MAX}}}$: the uncertainty is taken into account in the exponential function.

$$R_{TV} = C_t N_{PT}$$

$$\Rightarrow$$

$$\mathbb{E}[R_{TV}] = C_t \mathbb{E}[N_{PT}] = \begin{cases} NS * \mathcal{C} * C_t e^{-\frac{C_t}{C_{MAX}}}, & C_t \leq C_{MAX} \\ 0, & C_t > C_{MAX} \end{cases}$$

We want to underline again the ideality of this model and in particular of the discontinuity; nevertheless, the values far enough from the threshold are pretty interesting and satisfying.

Reliability engineering comparison

The theory of reliability and quality control is a branch of safety engineering which fits quite well our model. The main functions of this theory are reliability $R(t)$ and its opposite $F(t) = 1 - R(t)$. If we consider a large sample of industrial components N_0 and we split it in $N_f(t)$ and $N_b(t)$, the number of functioning and broken pieces, at the time t , we have:

$$R(t) = \frac{N_f(t)}{N_0}, F(t) = \frac{N_b(t)}{N_0}$$

From R we can define the failure rate, $\lambda(t) = -\frac{dR}{dt}/R$: if we consider a constant rate we get

$R(t) = e^{-\lambda t}, F(t) = 1 - e^{-\lambda t}$. In more complicated conditions, the general equation for R is:

$$\begin{cases} \frac{dR(t)}{dt} = \exp\left(-\left(\frac{t-\gamma}{\eta}\right)^\beta\right), & t \geq \gamma \\ R(t) = 1, & t < \gamma \end{cases}$$

This piecewise defined Weibull function takes into account γ , the minimum life of a component, η , the proper life, β , the Weibull shape parameter. We can observe that our reluctance model is quite similar to this framework, if we see the ticket cost as the time.

Now, let consider another point of view, in which the train resembles an electric circuit of NS (number of seats) parallel resistors. If a resistor stops working the remaining circuit keeps going on; we also have a constraint on the minimum power generated (total operating cost for a single run).

In order to balance the reduced number of resistors, the other components can undertake the total load if we enhance the voltage (C_t). But when the number of functioning pieces goes down a certain threshold, they cannot develop anymore the power we need. In our case when the required price/voltage increases too much and goes over C_{MAX} , the expected revenues $\mathbb{E}[R_{TV}]$ go to 0 (it represents the shutdown).

Financial Results

The last chapter of this analysis deals with the computation of HS train financial sustainability after the virus spread. We computed the expected value of revenues for a single train run on Rome-Milan and Turin-Milan paths: on an Excel sheet we calculated the revenues and net profit or loss in every condition, depending on different capacity, ticket cost and operating cost per kilometer.

Here we show all the tables of a *Frecciarossa 1000* (NS = 450) for $c = 1, 0.8, 0.5$. We estimated a ticket threshold of €160 and €75 and considered different values of m . In our ideas, $m = 18 \frac{\text{€}}{\text{km}}$ means that the company management is perfect, the state contribution is consistent and the national economic recovery is remarkable; on the other side, $m = 33 \frac{\text{€}}{\text{km}}$ shows a bad management, insufficient state incentives and no recovery.

TICKET PRICE	RUN REVENUES (C=100%)	REVENUES AT 80%	REVENUES AT 50%
50	16461,35165	13169,08132	8230,675826
60	18556,81053	14845,44842	9278,405264
70	20337,92858	16270,34287	10168,96429
80	21835,10375	17468,083	10917,55187
90	23076,2044	18460,96352	11538,1022
100	24086,76428	19269,41143	12043,38214
110	24890,16311	19912,13049	12445,08155
120	25507,79385	20406,23508	12753,89692
130	25959,21764	20767,37411	12979,60882
140	26262,30724	21009,84579	13131,15362
150	26433,3798	21146,70384	13216,6899
160	26487,31976	21189,85581	13243,65988
170	0	0	0

Table 6. Revenues for Rome-Milan, $C_{MAX} = \text{€}160$.

TICKET PRICE	RUN REVENUES (C=100%)	REVENUES AT 80%	REVENUES AT 50%
20	6893,355045	5514,684036	3446,677523
25	8060,977244	6448,781795	4030,488622
30	9049,320621	7239,456497	4524,660311
35	9876,653093	7901,322474	4938,326547
40	10559,63195	8447,705561	5279,815976
45	11113,43563	8890,748505	5556,717815
50	11551,88518	9241,508143	5775,942589
55	11887,5562	9510,044962	5943,778101
60	12131,88203	9705,505625	6065,941016
65	12295,24875	9836,198998	6147,624373
70	12387,08271	9909,666166	6193,541354
75	12415,93114	9932,744912	6207,96557
80	0	0	0

Table 9. Revenues for Turin-Milan, $C_{MAX} = \text{€}75$.

TICKET PRICE	m = 18 €/km			m = 21 €/km			m = 24 €/km			m = 27 €/km			m = 30 €/km			m = 33 €/km		
	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5
50	7641,352	4349,081	-589,324	6171,352	2879,081	-2059,32	4701,352	1409,081	-3529,32	3231,352	-60,9187	-4999,32	1761,352	-1530,92	-6469,32	291,3517	-3000,92	-7939,32
60	9736,811	6025,448	458,4053	8266,811	4555,448	-1011,59	6796,811	3085,448	-2481,59	5326,811	1615,448	-3951,59	3856,811	145,4484	-5421,59	2386,811	-1324,55	-6891,59
70	11517,93	7450,343	1348,964	10047,93	5980,343	-121,036	8577,929	4510,343	-1591,04	7107,929	3040,343	-3061,04	5637,929	1570,343	-4531,04	4167,929	100,3429	-6001,04
80	13015,1	8648,083	2097,552	11545,1	7178,083	627,5519	10075,1	5708,083	-842,448	8605,104	4238,083	-2312,45	7135,104	2768,083	-3782,45	5665,104	1298,083	-5252,45
90	14256,2	9640,964	2718,102	12786,2	8170,964	1248,102	11316,2	6700,964	-221,898	9846,204	5230,964	-1691,9	8376,204	3760,964	-3161,9	6906,204	2290,964	-4631,9
100	15266,76	10449,41	3223,382	13796,76	8979,411	1753,382	12326,76	7509,411	283,3821	10856,76	6039,411	-1186,62	9386,764	4569,411	-2656,62	7916,764	3099,411	-4126,62
110	16070,16	11092,13	3625,082	14600,16	9622,13	2155,082	13130,16	8152,13	685,0816	11660,16	6682,13	-784,918	10190,16	5212,13	-2254,92	8720,163	3742,13	-3724,92
120	16687,79	11586,24	3933,897	15217,79	10116,24	2463,897	13747,79	8646,235	993,8969	12277,79	7176,235	-476,103	10807,79	5706,235	-1946,1	9337,794	4236,235	-3416,1
130	17139,22	11947,37	4159,609	15669,22	10477,37	2689,609	14199,22	9007,374	1219,609	12729,22	7537,374	-250,391	11259,22	6067,374	-1720,39	9789,218	4597,374	-3190,39
140	17442,31	12189,85	4311,154	15972,31	10719,85	2841,154	14502,31	9249,846	1371,154	13032,31	7779,846	-98,8464	11562,31	6309,846	-1568,85	10092,31	4839,846	-3038,85
150	17613,38	12326,7	4396,69	16143,38	10856,7	2926,69	14673,38	9386,704	1456,69	13203,38	7916,704	-13,3101	11733,38	6446,704	-1483,31	10263,38	4976,704	-2953,31
160	17667,32	12369,86	4423,66	16197,32	10899,86	2953,66	14727,32	9429,856	1483,66	13257,32	7959,856	13,65988	11787,32	6489,856	-1456,34	10317,32	5019,856	-2926,34
170	-8820	-8820	-8820	-10290	-10290	-10290	-11760	-11760	-11760	-13230	-13230	-13230	-14700	-14700	-14700	-16170	-16170	-16170

Table 10. Net profit/loss for Rome-Milan, $l = 490$ km.

TICKET PRICE	m = 18 €/km			m = 21 €/km			m = 24 €/km			m = 27 €/km			m = 30 €/km			m = 33 €/km		
	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5	C = 1	C = 0,8	C = 0,5
20	4229,355	2850,684	782,6775	3785,355	2406,684	338,6775	3341,355	1962,684	-105,322	2897,355	1518,684	-549,322	2453,355	1074,684	-993,322	2009,355	630,684	-1437,32
25	5396,977	3784,782	1366,489	4952,977	3340,782	922,4886	4508,977	2896,782	478,4886	4064,977	2452,782	34,48862	3620,977	2008,782	-409,511	3176,977	1564,782	-853,511
30	6385,321	4575,456	1860,66	5941,321	4131,456	1416,66	5497,321	3687,456	972,6603	5053,321	3243,456	528,6603	4609,321	2799,456	84,66031	4165,321	2355,456	-359,34
35	7212,653	5237,322	2274,327	6768,653	4793,322	1830,327	6324,653	4349,322	1386,327	5880,653	3905,322	942,3265	5436,653	3461,322	498,3265	4992,653	3017,322	54,32655
40	7895,632	5783,706	2615,816	7451,632	5339,706	2171,816	7007,632	4895,706	1727,816	6563,632	4451,706	1283,816	6119,632	4007,706	839,816	5675,632	3563,706	395,816
45	8449,436	6226,749	2892,718	8005,436	5782,749	2448,718	7561,436	5338,749	2004,718	7117,436	4894,749	1560,718	6673,436	4450,749	1116,718	6229,436	4006,749	672,7178
50	8887,885	6577,508	3111,943	8443,885	6133,508	2667,943	7999,885	5689,508	2223,943	7555,885	5245,508	1779,943	7111,885	4801,508	1335,943	6667,885	4357,508	891,9426
55	9223,556	6846,045	3279,778	8779,556	6402,045	2835,778	8335,556	5958,045	2391,778	7891,556	5514,045	1947,778	7447,556	5070,045	1503,778	7003,556	4626,045	1059,778
60	9467,882	7041,506	3401,941	9023,882	6597,506	2957,941	8579,882	6153,506	2513,941	8135,882	5709,506	2069,941	7691,882	5265,506	1625,941	7247,882	4821,506	1181,941
65	9631,249	7172,199	3483,624	9187,249	6728,199	3039,624	8743,249	6284,199	2595,624	8299,249	5840,199	2151,624	7855,249	5396,199	1707,624	7411,249	4952,199	1263,624
70	9723,083	7245,666	3529,541	9279,083	6801,666	3085,541	8835,083	6357,666	2641,541	8391,083	5913,666	2197,541	7947,083	5469,666	1753,541	7503,083	5025,666	1309,541
75	9751,931	7268,745	3543,966	9307,931	6824,745	3099,966	8863,931	6380,745	2655,966	8419,931	5936,745	2211,966	7975,931	5492,745	1767,966	7531,931	5048,745	1323,966
80	-2664	-2664	-2664	-3108	-3108	-3108	-3552	-3552	-3552	-3996	-3996	-3996	-4440	-4440	-4440	-4884	-4884	-4884

Table 11. Net profit/loss for Turin-Milan, $l = 148$ km.

We chose to represent in red cells the negative values, which correspond to losses. Green and yellow cells represent instead positive values of profit. In particular, we decided to highlight in yellow those runs for which the expected value of profit we predicted was smaller than €2000 for Rome-Milan and €1000 for Turin-Milan: sustainability analysis have to take into account that revenues magnitude needs dramatic profits to allow the train to undertake the run. In fact, since we are considering expected values, we cannot be sure that a positive but small profit will occur.

At first sight, Turin-Milan could be much more convenient for the company, because of the significant number of green cells: even with low capacity and low-ticket price, we could get adequate profits. A possible explanation lies in economy of scale, since on short distances the operating costs have a lower impact than the fixed costs.

A deeper analysis shows this to be a biased result. The first reason is the real total path of the train: a *Frecciarossa 1000* travelling in this direction keeps running at least until Rome and often reaches even Naples and Salerno, generating huge costs. The second reason lies hidden in the computation of m : this coefficient is a consequence of the impact of HS on the Balance Sheet and this kind of train never stops after such short distances.

In conclusion, we are quite confident that the halved capacity implies the economical unsustainability of HS railway transport. Our analysis could be even supported by Italo-NTV decision to completely stop the services, a fortiori because their data analysis and models are many times more effective and accurate than ours.

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